

WILLIAM GALVIN

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EDUCATION

Honours Bachelor of Computer Science (Mobile Computing), Sheridan College

2023 – Dec. 2027

SKILLS

Languages: C++, Python, Java, C, TypeScript

Technologies: Linux, kdb+/q, Polars, SQL, CMake/Ninja, Qt, LLVM, Git, Docker, Bash

Concepts: Low-Latency Systems, Concurrency, Compiler Design, Systems Programming, Networking

EXPERIENCE

Software Developer Intern, Quantitative Technology Services

May – Aug. 2026

RBC Capital Markets, Central Risk Book

Toronto, ON

- Built an intraday **fill reconciliation** monitor in Polars over two **kdb+** tables, matching executed fills from the desk's HFT system against the drop copies recorded by the strategy system that instructed them. Flags and localizes one-sided breaks **within 2 minutes**, replacing manual discovery and cutting the time to get both systems back in sync.
- Wrote a **Qt/C++ launcher** replacing manual deployment of the trading GUI that **controls live market-making strategies**: resolves the latest release from the GitHub Enterprise API, downloading uncached versions — with support for pinned rollbacks and opt-in prereleases. Authored the GUI's **CMake build system** as a shared-dependency monorepo with full unit test coverage and documentation.
- Rewrote a legacy **Java** DMM compliance reporter against revised requirements, replacing scratch-table repopulation and a monolithic query with a dataframe pipeline that normalizes exchange-published files and database sources independently before joining into a persistent securities table — reducing report generation to a single join against the security master.
- Cut trading-system logs **20.9%** beyond gzip across a seven-year retention archive using frequency-analysis key substitution, keeping output human-readable for incident triage and byte-exact on rehydration.

Lead Organizer & Founder

November 2025

Sheridan Datathon

Mississauga, ON

- Founded Sheridan's first datathon, themed on UN Sustainable Development Goals, after being brought in by the Google Developer Group; drew **200+ participants** and now runs annually under a successor team.
- Led **30+ organizers** and **9 mentors**, secured **\$6.8k** in sponsorships, and owned end-to-end execution: venue, competition design, website, judging, marketing, and registration.

PROJECTS

Low-Latency Benchmarking Engine

C++23, CMake, Python

- Built a cycle-accurate benchmarking framework using RDTSC, with a generic **stats::Summary<T>** computing latency distributions (min/max/percentiles/stddev) across runs and a CMake/Ninja build targeting reproducible, zero-overhead measurement.
- Used it to benchmark a policy-based ring buffer, tuning the design against measured tail latency rather than throughput averages.

Cherry Compiler

C++, LLVM

- Long-running compiler project for a dual-language system — Pit (low-level) and Cherry (high-level, compiling to Pit) — with hand-written lexers and recursive descent parsers against formally specified grammars.
- Currently lowering the AST into a custom IR and emitting LLVM IR from it.